1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027)

26th & 27th February , 2027

Department of Electrical & Computer Engineering

&

Department of Electrical Engineering

Ajay Binay Institute Of Technology (Autonomous), Kataka

About SAGAR2027 :

The **1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy** (**SAGAR2027**) will be held on **February 26–27, 2027**, at **ABIT(Autonomous)** in **hybrid mode**. The conference is jointly organized by the **Department of Electrical and Computer Engineering** and **Department of Electrical Engineering.**

SAGAR2027 focuses on the application of **soft computing techniques which includes artificial intelligence, machine learning, neural networks, fuzzy logic, evolutionary algorithms, optimization, data science,** and **analytics that address challenges in smart grids and renewable energy systems.**

The conference aims to bring together researchers, academicians, industry professionals, and students to discuss recent advances in intelligent technologies for **power systems, renewable energy integration, energy management, forecasting, control, optimization,** and **reliability enhancement**. It provides a platform for knowledge sharing, collaboration, and exploration of future trends in sustainable and smart energy systems.

About Ajay Binay Institute of Technology (ABIT) Autonomous college :

Ajay Binay Institute of Technology (ABIT) is approved by **AICTE**, affiliated with **BPUT**, and has been granted **Autonomous Status by UGC**. ABIT group operates an **Engineering College** and an **Industrial Training Institute** in Cuttack, Odisha.

The group was established in **1993** with the aim to improve the availability of **skilled technical manpower** for industries as well as to enhance reach of technical education to rural Odisha.

The Institute is approved by **AICTE, New Delhi**, affiliated to **Biju Patnaik University of Technology, Rourkela** and **accredited by NAAC**. ABIT has a balanced distribution of technical streams with equal emphasis on the **core sector** and **IT sector**.

Important Date :

- **Submission Open: July 20,2026**
- **Submission Deadline: Nov 20, 2026**
- **Acceptance Notification: Jan 15, 2027**
- **Author Registration: Jan 27, 2027**
- **Camera Ready Paper Submission: Jan 25, 2027**
- **Conference dates: February 26-27, 2027**

Registration fee		
Category	**Indian Author**	**Foreign Author**
Full Registration	**10000 INR**	**200 USD**
Praticipation Fee	**2000 INR**	**50 USD**

*Note:

- **Registration is valid for UG/PG students and PhD scholar with valid Id card, Academician and Industry professionals.**
- **At least one author registration is mandatory for each accepted paper.**
- **The manuscript may contain 6 pages upto one additional page is permissible with Overlength page charge 1000 INR for Indian and USD 25 for foreign Author.**
- **Co-authors who wish to attend the conference in offline mode must register under the Attendee category.**

***All the presented & accepted papers will be considered for publication in Bentham Science Book Series (Scopus Indexed).**

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SAGAR2027 Website

https://www.sagar2027.com/

For Paper Submission

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TECHNICAL TRACKS:

Track 1: Power Systems and Smart Grids

- Power generation, transmission, and distribution
- Power system operation and control
- Smart grid technologies and applications
- Renewable energy integration and grid integration challenges
- Grid resilience and reliability enhancement
- Energy storage systems and management
- Demand-side management and demand response strategies
- Smart Grid Planning and Operation

Track 2: AI & ML application and Cyber security

- AI-Based Intrusion Detection Systems (IDS)
- Deep Learning Applications
- Phishing and Spam Detection Using Machine Learning
- Malware Classification with Deep Learning
- Network Anomaly Detection Using AI
- Adversarial Attacks and Defenses in Machine Learning

Track 3: Application of optimization techniques

- power system optimization
- Vehicle Routing and Logistics Optimization
- Hyperparameter Optimization in Machine Learning
- Smart Grid Energy Management Optimization
- Production and Inventory Scheduling Optimization
- Resource Allocation in Cloud Computing Using Optimization Algorithms

Track 4: Power Electronics and Intelligent Control Systems

- Intelligent control of power converters
- Advanced inverter technologies
- Control strategies for renewable energy systems
- FACTS devices and smart grid applications
- Adaptive and robust control techniques
- Embedded systems for smart energy applications
- Automation in power systems
- Electric Vehicle and Charging station

Track 5: Emerging Technologies and Future Trends

- Internet of Things (IoT) and communication techniques
- Quantum computing applications in energy systems
- Green hydrogen technologies
- AI-driven carbon neutrality solutions
- Internet of Energy (IoE)
- Smart energy markets and trading
- Digital transformation in power systems
- Future challenges in renewable energy and smart grids

Paper Submission Link: **https://cmt3.research.microsoft.com/SAGAR2027**

Tourist Attractions:

- **Barabati Stadium**
- **Silvercity Boat Club**
- **Shree Jagannath Temple, Puri**
- **Konark Sun Temple**
- **Barabati Fort**
- **Odisha State Maritime Museum**
- **Netaji Birth Place Museum**
- **Kataka Chandi Temple**
- **Chilika Lake**
- **Lingaraj Temple**
- **Dhauli santi stupa**
- **Khandagiri and Udayagiri**
- **Puri Sea Beach**



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